Qualification Pack

Drone Application Developer

QP Code: NIE/ELE/Q2501

Version: 1.0

NSQF Level: 5

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Contents

NIE/ELE/Q2501: Drone Application Developer

Brief Job Description

This comprehensive course on drone technology equips participants with the knowledge and practical skills needed to operate, assemble, and maintain drones effectively. The curriculum covers foundational concepts of drone systems, regulations, and ethical considerations, as well as advanced topics like subsystem integration, programming, and real-world applications. Participants will gain hands-on experience with tools and technologies such as Pixhawk flight controllers, Raspberry Pi, and Dronekit, ensuring proficiency in safe drone operation, troubleshooting, and maintenance for various industries.

Personal Attributes

The qualification requires individuals to possess a keen interest in drone technology and its applications, a detail-oriented mindset for troubleshooting and maintenance, and a strong sense of responsibility to adhere to safety and ethical standards in drone operations.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. NIE/ELE/N2507: Drone Technology, Regulations & Ethical Considerations-
2. NIE/ELE/N2508: Drone Subsystem-
3. NIE/ELE/N2503: Drone Fight Controller & Peripherals
4. NIE/ELE/N2504: Drone Payload Installation and Integration Testing
5. NIE/ELE/N2505: Drone Programming and its Applications
6. NIE/ELE/N2506: Drone Equipment Safety and Maintenance
7. DGT/VSQ/N0102: Employability Skills (60 Hours)
8. NIE/ELE/N2509: Implementation of Multi-Functional Drone Systems Using Pixhawk and Python Programming

Qualification Pack (QP) Parameters

Sector	Electronics
Sub-Sector	Applied Electronics

Qualification Pack

Occupation	Drone Technology
Country	India
NSQF Level	5
Credits	18
Aligned to NCO/ISCO/ISIC Code	5
Minimum Educational Qualification & Experience	UG in any field (Completed UG Diploma in Electronics and Communication Engineering/ Electrical Engineering/CS/IT and allied branches) with NA of experience NA OR Diploma (Completed 3 Years of Diploma in Electronics and Communication Engineering/ Electrical Engineering/CS/IT and allied branches after class 10th) with 1.5 years of experience OR Completed 2nd year diploma after 12th (in Electronics and Communication Engineering/ Electrical Engineering/CS/IT and allied branches) with NA of experience OR Previous relevant Qualification of NSQF Level (Acquired NSQF Level 4.5 in Electronics and Communication Engineering/ Electrical Engineering/CS/IT and allied branches) with 1.5 years of experience OR Previous relevant Qualification of NSQF Level (Acquired NSQF Level 4 in Electronics and Communication Engineering/ Electrical Engineering/CS/IT and allied branches) with 3 Years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	20 Years
Last Reviewed On	NA
Next Review Date	30/05/2027
NSQC Approval Date	30/05/2024
Version	1.0

Reference code on NQR	QG-05-EH-02596-2024-V1-NIELIT
NQR Version	1

NIE/ELE/N2507: Drone Technology, Regulations & Ethical Considerations-

Description

This course provides a comprehensive overview of drone technology, covering key topics such as the components and classifications of drones, principles of aerodynamics and flight mechanics, and the impact of weather on operations. Participants will gain insights into regulatory frameworks, with a focus on India, as well as ethical considerations like privacy and legal implications. The course also explores emerging trends and advancements in drone technology, preparing learners to navigate the technical, legal, and ethical aspects of this rapidly evolving field.

Scope

The scope covers the following :

- This course on Drone Technology, Regulations & Ethical Considerations provides a comprehensive overview of the technical aspects, regulatory compliance, and ethical issues surrounding drone use. Participants will gain insights into drone mechanics, flight dynamics, and weather impacts, alongside a deep dive into the legal frameworks governing drone operations, particularly focusing on India. The curriculum also addresses the ethical considerations and privacy concerns of drone technology, preparing students for a variety of roles in drone operation, policy development, or tech innovation. The course prepares learners to effectively navigate the complexities of this rapidly evolving field, highlighting future trends and opportunities for advancement.

Elements and Performance Criteria

Understanding Drone Technology:

To be competent, the user/individual on the job must be able to:

PC1. Successfully define and classify drones, demonstrating knowledge of terminology.

PC2. Identify and describe drone components accurately, showing understanding of their functions.

PC3. Explain principles of aerodynamics and flight mechanics in a clear and coherent manner.

Comprehending Drone Classification:

To be competent, the user/individual on the job must be able to:

PC4. Classify drones effectively based on design, purpose, and size, demonstrating clarity and accuracy.

PC5. Differentiate drones based on autonomy and payload capabilities with precision.

PC6. Discuss the importance of materials and size variations in drone construction, providing relevant examples.

Navigating Drone Regulations and Ethics:

To be competent, the user/individual on the job must be able to:

PC7. Interpret and apply drone regulations correctly in various scenarios, demonstrating adherence to legal requirements.

PC8. Navigate the certification process for drone operation in India successfully, completing required procedures accurately.

PC9. Analyze ethical considerations and privacy concerns related to drone usage, offering insightful perspectives and recommendations.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Gain a thorough understanding of the key components of drones, including airframes, propulsion systems, and avionics, and learn how these elements contribute to drone flight and stability.

KU2. Develop insights into the regulatory landscapes governing drone operations, with an emphasis on compliance and ethical considerations, including privacy and data security.

KU3. Explore current and future uses of drone technology across various sectors and understand the latest advancements, such as autonomous drones and AI integration, shaping the field.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Participants will learn to effectively operate and manage drone systems, honing skills in navigation, flight planning, and safety protocols to ensure efficient and secure drone operations.

GS2. Acquire the skills necessary to navigate complex regulatory environments, understand certification processes, and ensure adherence to all legal and ethical standards associated with drone operations.

GS3. Encourage creative thinking and problem-solving skills through real-world scenarios and case studies, enabling participants to devise innovative solutions for integrating drone technology into various business and environmental challenges.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Understanding Drone Technology:	**11**	**5**	**-**	**-**
PC1. Successfully define and classify drones, demonstrating knowledge of terminology.	-	-	-	-
PC2. Identify and describe drone components accurately, showing understanding of their functions.	-	-	-	-
PC3. Explain principles of aerodynamics and flight mechanics in a clear and coherent manner.	-	-	-	-
Comprehending Drone Classification:	**11**	**5**	**-**	**-**
PC4. Classify drones effectively based on design, purpose, and size, demonstrating clarity and accuracy.	-	-	-	-
PC5. Differentiate drones based on autonomy and payload capabilities with precision.	-	-	-	-
PC6. Discuss the importance of materials and size variations in drone construction, providing relevant examples.	-	-	-	-
Navigating Drone Regulations and Ethics:	**12**	**5**	**-**	**-**
PC7. Interpret and apply drone regulations correctly in various scenarios, demonstrating adherence to legal requirements.	-	-	-	-
PC8. Navigate the certification process for drone operation in India successfully, completing required procedures accurately.	-	-	-	-
PC9. Analyze ethical considerations and privacy concerns related to drone usage, offering insightful perspectives and recommendations.	-	-	-	-
NOS Total	**34**	**15**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N2507
NOS Name	Drone Technology, Regulations & Ethical Considerations-
Sector	Electronics
Sub-Sector	
Occupation	Drone Technology
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

NIE/ELE/N2508: Drone Subsystem-

Description

This course offers an in-depth understanding of the essential components of drone systems, including power, propulsion, and flight control mechanisms. Participants will explore navigation systems like GPS, communication tools, and sensors such as cameras and LiDAR, along with data processing and storage techniques. The curriculum also addresses structural elements, safety features, and redundancy systems, providing a solid foundation for designing and managing drone subsystems.

Scope

The scope covers the following :

- The course equips participants with comprehensive knowledge of drone subsystems, preparing them for roles in design, integration, and maintenance of UAV technology. It covers critical aspects such as power management, propulsion, navigation, communication, and sensor integration, along with safety and redundancy systems. This broad scope enables learners to apply their skills in industries like logistics, agriculture, surveillance, and disaster management, fostering innovation in drone technology applications.

Elements and Performance Criteria

Understanding Drone Subsystems:

To be competent, the user/individual on the job must be able to:

PC1. Demonstrate comprehensive knowledge of each drone subsystem, including their components and functionalities, through written or verbal assessments.

PC2. Successfully explain the interrelationship between different subsystems and how they collectively contribute to overall drone operation and performance.

PC3. Present clear and coherent explanations of complex subsystem concepts, facilitating understanding among peers and instructors.

Analyzing Subsystem Components:

To be competent, the user/individual on the job must be able to:

PC4. Identify and describe the key components of each subsystem accurately, demonstrating a deep understanding of their roles and technical specifications.

PC5. Analyze the performance characteristics of subsystem components effectively, evaluating factors such as power consumption, thrust, data processing speed, etc., to assess their suitability for specific drone applications.

PC6. Discuss the importance of redundancy and safety systems in drone subsystems, highlighting their significance in ensuring reliability and mitigating potential failures.

Troubleshooting and Maintenance

To be competent, the user/individual on the job must be able to:

PC7. Successfully diagnose and resolve simulated or real-world issues related to drone subsystems within a given timeframe, employing appropriate problem-solving strategies and tools.

PC8. Perform routine maintenance tasks for each subsystem competently, following established procedures and safety guidelines to prevent accidents or damage.

PC9. Implement effective communication and collaboration skills when working in teams to troubleshoot complex subsystem issues or perform maintenance tasks, demonstrating professionalism and teamwork.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Gain a thorough understanding of drone components such as power systems, propulsion mechanisms, flight control, navigation, and communication technologies, along with their integration for optimal performance.

KU2. Develop expertise in deploying advanced sensors like LiDAR, infrared, and ultrasonic systems, as well as onboard data processing and storage techniques to enhance drone functionality.

KU3. Acquire practical knowledge of drone structural design, safety measures, and redundancy systems, alongside their applications in industries like agriculture, defense, logistics, and environmental monitoring.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Develop the ability to analyze, assemble, and troubleshoot various drone subsystems, including propulsion, navigation, and communication systems, ensuring optimal functionality and performance.

GS2. Enhance problem-solving abilities to design and implement customized drone solutions tailored to specific industry needs, fostering innovation in applications such as surveying, delivery, and surveillance.

GS3. Gain professional proficiency in ensuring drone safety, implementing redundancy measures, and adhering to regulatory and ethical standards for efficient and responsible drone operations.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Understanding Drone Subsystems:	**11**	**5**	**-**	**-**
PC1. Demonstrate comprehensive knowledge of each drone subsystem, including their components and functionalities, through written or verbal assessments.	-	-	-	-
PC2. Successfully explain the interrelationship between different subsystems and how they collectively contribute to overall drone operation and performance.	-	-	-	-
PC3. Present clear and coherent explanations of complex subsystem concepts, facilitating understanding among peers and instructors.	-	-	-	-
Analyzing Subsystem Components:	**11**	**5**	**-**	**-**
PC4. Identify and describe the key components of each subsystem accurately, demonstrating a deep understanding of their roles and technical specifications.	-	-	-	-
PC5. Analyze the performance characteristics of subsystem components effectively, evaluating factors such as power consumption, thrust, data processing speed, etc., to assess their suitability for specific drone applications.	-	-	-	-
PC6. Discuss the importance of redundancy and safety systems in drone subsystems, highlighting their significance in ensuring reliability and mitigating potential failures.	-	-	-	-
Troubleshooting and Maintenance	**11**	**5**	**-**	**-**
PC7. Successfully diagnose and resolve simulated or real-world issues related to drone subsystems within a given timeframe, employing appropriate problem-solving strategies and tools.	-	-	-	-
PC8. Perform routine maintenance tasks for each subsystem competently, following established procedures and safety guidelines to prevent accidents or damage.	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. Implement effective communication and collaboration skills when working in teams to troubleshoot complex subsystem issues or perform maintenance tasks, demonstrating professionalism and teamwork.	-	-	-	-
NOS Total	**33**	**15**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N2508
NOS Name	Drone Subsystem-
Sector	Electronics
Sub-Sector	
Occupation	Drone Technology
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

NIE/ELE/N2503: Drone Fight Controller & Peripherals

Description

This course provides both foundational and advanced knowledge in drone flight controller technology, emphasizing microcontroller and embedded systems, Pixhawk architecture, and signal processing methods. Learners will gain hands-on experience in configuring essential drone peripherals, including radio controllers, GPS modules, and Electronic Speed Controllers (ESCs).

Scope

The scope covers the following :

- The qualification equips learners with the technical knowledge and practical skills required to operate, configure, and integrate drone flight controllers and peripherals. It covers critical aspects of microcontroller systems, Pixhawk architecture, and peripheral setup, including GPS modules, radio controllers, and ESCs. This program prepares individuals for roles in drone assembly, maintenance, and operation across industries such as logistics, agriculture, surveillance, and environmental monitoring, ensuring they meet the growing demands of the UAV sector.

Elements and Performance Criteria

Understanding and Configuring Drone Flight Controllers:

To be competent, the user/individual on the job must be able to:

PC1. Demonstrate a clear understanding of microcontroller and embedded systems concepts related to drone operations.

PC2. Explain the architecture and components of the Pixhawk flight controller and its role in managing drone functions

PC3. Identify and process PWM and PPM signals effectively.

Integration and Calibration of Key Components:

To be competent, the user/individual on the job must be able to:

PC4. Set up and configure the radio controller to establish seamless communication with the flight controller.

PC5. Connect and calibrate GPS modules, ensuring accurate navigation and system synchronization.

PC6. Integrate and calibrate ESCs for stable propulsion system performance.

Assembly and Power System Integration:

To be competent, the user/individual on the job must be able to:

PC7. Safely attach the battery to the drone frame, ensuring proper wiring and secure mounting.

PC8. Assemble the flight controller and peripheral components into the drone structure with precision.

PC9. Verify the assembly and power system integration for operational readiness.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the architecture, functionality, and configuration of flight controllers, including microcontroller principles and Pixhawk architecture, essential for drone operations.

KU2. Gain knowledge of PWM and PPM signals, their role in drone control, and the integration of key peripherals such as GPS modules, radio controllers, and ESCs for seamless functionality.

KU3. Develop an understanding of the assembly process, structural integrity, and adherence to safety standards, ensuring reliable and secure drone operations.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Develop hands-on skills in assembling drone components, integrating peripherals, and troubleshooting flight controller systems for efficient and reliable performance.

GS2. Demonstrate proficiency in configuring flight controllers, calibrating GPS modules, ESCs, and other peripherals to ensure seamless drone functionality.

GS3. Apply industry-standard safety practices during assembly and operation, ensuring compliance with legal regulations and optimizing drone system reliability for professional applications.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Understanding and Configuring Drone Flight Controllers:	**11**	**5**	**-**	**-**
PC1. Demonstrate a clear understanding of microcontroller and embedded systems concepts related to drone operations.	-	-	-	-
PC2. Explain the architecture and components of the Pixhawk flight controller and its role in managing drone functions	-	-	-	-
PC3. Identify and process PWM and PPM signals effectively.	-	-	-	-
Integration and Calibration of Key Components:	**11**	**5**	**-**	**-**
PC4. Set up and configure the radio controller to establish seamless communication with the flight controller.	-	-	-	-
PC5. Connect and calibrate GPS modules, ensuring accurate navigation and system synchronization.	-	-	-	-
PC6. Integrate and calibrate ESCs for stable propulsion system performance.	-	-	-	-
Assembly and Power System Integration:	**11**	**5**	**-**	**-**
PC7. Safely attach the battery to the drone frame, ensuring proper wiring and secure mounting.	-	-	-	-
PC8. Assemble the flight controller and peripheral components into the drone structure with precision.	-	-	-	-
PC9. Verify the assembly and power system integration for operational readiness.	-	-	-	-
NOS Total	**33**	**15**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N2503
NOS Name	Drone Fight Controller & Peripherals
Sector	Electronics
Sub-Sector	
Occupation	Drone Technology
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

NIE/ELE/N2504: Drone Payload Installation and Integration Testing

Description

This qualification provides comprehensive training in drone payload installation and integration, combining firmware programming with hardware assembly. Participants will gain hands-on experience in configuring and flashing ArduPilot firmware to the Pixhawk controller, assembling drone frames, integrating sensors, and calibrating propulsion systems for optimal performance. The course emphasizes safety protocols, legal compliance, and flight testing, equipping learners with the skills needed for professional roles in drone system integration and deployment across diverse industries.

Scope

The scope covers the following :

- The qualification equips learners with the knowledge and skills required for drone payload installation and integration testing. It covers firmware programming, including configuring and flashing ArduPilot firmware, as well as hardware assembly processes such as frame construction, sensor integration, and propulsion system calibration. Emphasizing safety protocols, legal compliance, and flight testing, this qualification prepares individuals to excel in roles involving drone system integration, testing, and deployment across industries such as agriculture, logistics, surveillance, and environmental monitoring.

Elements and Performance Criteria

Firmware Management and Programming:

To be competent, the user/individual on the job must be able to:

PC1. Understanding the process of firmware upgrading and programming for the Pixhawk flight controller, specifically focusing on porting ArduPilot firmware and configuring settings.

PC2. Proficiency in downloading, flashing, and verifying ArduPilot firmware onto the Pixhawk flight controller to ensure compatibility and functionality.

PC3. Skill in troubleshooting firmware-related issues and implementing corrective actions to ensure successful firmware operation.

Assembly, Integration, and Testing:

To be competent, the user/individual on the job must be able to:

PC4. Competence in assembling drone frames and mounts, including soldering electronics components and devices as per specifications.

PC5. Ability to integrate sensors, receivers, and propulsion systems into the drone frame, ensuring proper wiring and connectivity.

PC6. Proficiency in conducting calibration procedures for ESCs, attaching propellers, and performing safety and legal checks before testing the drone's flight capabilities.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the principles of firmware programming, including porting, configuring, and flashing ArduPilot firmware to the Pixhawk flight controller, ensuring optimal drone functionality.

KU2. Gain comprehensive knowledge of assembling drone frames, soldering electronic components, integrating sensors, receivers, and propulsion systems, and calibrating ESCs for seamless operation.

KU3. Understand safety protocols, legal considerations, and testing procedures essential for reliable and compliant drone operations, including pre-flight inspections and troubleshooting techniques.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Develop practical expertise in assembling drone components, integrating payloads, and calibrating systems to ensure seamless operation and reliability.

GS2. Demonstrate proficiency in identifying and resolving technical issues during firmware programming, hardware assembly, and integration testing, ensuring optimal drone performance.

GS3. Apply industry-standard safety practices and comply with legal regulations during drone assembly, testing, and deployment, preparing for professional roles in the UAV industry.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Firmware Management and Programming:	**17**	**8**	**-**	**-**
PC1. Understanding the process of firmware upgrading and programming for the Pixhawk flight controller, specifically focusing on porting ArduPilot firmware and configuring settings.	-	-	-	-
PC2. Proficiency in downloading, flashing, and verifying ArduPilot firmware onto the Pixhawk flight controller to ensure compatibility and functionality.	-	-	-	-
PC3. Skill in troubleshooting firmware-related issues and implementing corrective actions to ensure successful firmware operation.	-	-	-	-
Assembly, Integration, and Testing:	**16**	**7**	**-**	**-**
PC4. Competence in assembling drone frames and mounts, including soldering electronics components and devices as per specifications.	-	-	-	-
PC5. Ability to integrate sensors, receivers, and propulsion systems into the drone frame, ensuring proper wiring and connectivity.	-	-	-	-
PC6. Proficiency in conducting calibration procedures for ESCs, attaching propellers, and performing safety and legal checks before testing the drone's flight capabilities.	-	-	-	-
NOS Total	**33**	**15**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N2504
NOS Name	Drone Payload Installation and Integration Testing
Sector	Electronics
Sub-Sector	
Occupation	Drone Technology
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

NIE/ELE/N2505: Drone Programming and its Applications

Description

This qualification provides hands-on training in drone programming and applications, focusing on Python programming and hardware integration. Participants will learn essential Python concepts tailored for drone operations, set up development environments with Raspberry Pi and Pixhawk, and configure tools like Dronekit and MAVProxy. The course explores real-world applications, including agriculture, infrastructure inspection, environmental monitoring, search and rescue, and thermal imaging, equipping learners with the skills to program and deploy drones across various industries.

Scope

The scope covers the following :

- This qualification enables learners to program and integrate drones for a variety of industrial applications. It includes Python programming, hardware setup with Raspberry Pi and Pixhawk, and the use of tools like Dronekit and MAVProxy. With insights into practical applications such as agriculture, infrastructure inspection, environmental monitoring, and search and rescue, the qualification prepares individuals for roles in drone programming, integration, and deployment across multiple sectors.

Elements and Performance Criteria

Python Programming for Drone Applications:

To be competent, the user/individual on the job must be able to:

PC1. Successfully demonstrate proficiency in Python syntax, data types, and control structures through coding exercises and assignments.

PC2. Develop functional Python programs using advanced concepts like functions, modules, file handling, exceptions handling, and classes to address drone-related tasks and challenges.

PC3. Apply Python programming skills to create innovative solutions for drone applications, showcasing creativity and problem-solving abilities.

Setting up Host Environment and Hardware Integration:

To be competent, the user/individual on the job must be able to:

PC4. Configure Raspberry Pi accurately, demonstrating understanding of its architecture and pin description.

PC5. Successfully mount Raspbian OS on an SD card and boot and configure the Raspberry Pi for integration with Pixhawk hardware.

PC6. Establish effective communication and compatibility between Raspberry Pi and Pixhawk hardware, ensuring seamless integration for programming and controlling the drone.

Drone Applications and Case Studies:

To be competent, the user/individual on the job must be able to:

PC7. Analyze drone applications in various fields and identify suitable solutions for specific use cases, demonstrating critical thinking and domain knowledge.

PC8. Evaluate the feasibility and effectiveness of drone applications through case studies and propose appropriate strategies to address challenges and achieve desired outcomes.

PC9. Design and execute case study-based projects to demonstrate practical implementation of drone applications, showcasing understanding and proficiency in utilizing drones for real-world scenarios.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the fundamentals of Python programming, including flow control, functions, file handling, and exception management, tailored specifically for drone applications.

KU2. Gain comprehensive knowledge of setting up and configuring hardware, including Raspberry Pi, Pixhawk, and essential tools like Dronekit and MAVProxy, for seamless drone operation.

KU3. Develop an understanding of the diverse applications of drones in fields such as agriculture, infrastructure inspection, environmental monitoring, search and rescue, and thermal imaging.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Develop practical expertise in Python programming and configuring development environments, enabling precise control and customization of drone operations.

GS2. Acquire hands-on skills in integrating and troubleshooting drone hardware components, including Raspberry Pi, Pixhawk, and related tools, ensuring reliable and efficient drone functionality.

GS3. Demonstrate the ability to apply drone technology effectively in real-world scenarios, such as agriculture, infrastructure inspection, and search and rescue, addressing industry-specific challenges with innovative solutions.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Python Programming for Drone Applications:	**12**	**5**	**-**	**-**
PC1. Successfully demonstrate proficiency in Python syntax, data types, and control structures through coding exercises and assignments.	-	-	-	-
PC2. Develop functional Python programs using advanced concepts like functions, modules, file handling, exceptions handling, and classes to address drone-related tasks and challenges.	-	-	-	-
PC3. Apply Python programming skills to create innovative solutions for drone applications, showcasing creativity and problem-solving abilities.	-	-	-	-
Setting up Host Environment and Hardware Integration:	**11**	**5**	**-**	**-**
PC4. Configure Raspberry Pi accurately, demonstrating understanding of its architecture and pin description.	-	-	-	-
PC5. Successfully mount Raspbian OS on an SD card and boot and configure the Raspberry Pi for integration with Pixhawk hardware.	-	-	-	-
PC6. Establish effective communication and compatibility between Raspberry Pi and Pixhawk hardware, ensuring seamless integration for programming and controlling the drone.	-	-	-	-
Drone Applications and Case Studies:	**11**	**5**	**-**	**-**
PC7. Analyze drone applications in various fields and identify suitable solutions for specific use cases, demonstrating critical thinking and domain knowledge.	-	-	-	-
PC8. Evaluate the feasibility and effectiveness of drone applications through case studies and propose appropriate strategies to address challenges and achieve desired outcomes.	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. Design and execute case study-based projects to demonstrate practical implementation of drone applications, showcasing understanding and proficiency in utilizing drones for real-world scenarios.	-	-	-	-
NOS Total	**34**	**15**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N2505
NOS Name	Drone Programming and its Applications
Sector	Electronics
Sub-Sector	
Occupation	Drone Technology
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

NIE/ELE/N2506: Drone Equipment Safety and Maintenance

Description

This qualification focuses on drone safety and maintenance, equipping participants with the skills to ensure reliability and performance. It covers pre-flight inspections, battery management, propulsion system upkeep, and calibration of flight controllers, navigation systems, and cameras. Participants will also learn software updates, flight log analysis, and troubleshooting techniques for handling emergencies. This program prepares individuals to maintain and optimize drones for safe and efficient use in diverse applications.

Scope

The scope covers the following :

- This qualification prepares individuals to ensure the safe and efficient operation of drones through comprehensive maintenance and troubleshooting. It includes pre-flight inspections, battery and propulsion management, and calibration of key components like flight controllers and navigation systems. Participants will also gain skills in software updates, flight log analysis, and handling emergencies, making it applicable across industries such as agriculture, surveillance, infrastructure inspection, and disaster response.

Elements and Performance Criteria

Drone Maintenance and Troubleshooting:

To be competent, the user/individual on the job must be able to:

PC1. Demonstrate understanding of the importance of maintenance and troubleshooting through written assessments or discussions.

PC2. Apply safety considerations and best practices learned to perform maintenance tasks and troubleshoot common drone issues effectively.

Equipment Inspection and Maintenance Procedures:

To be competent, the user/individual on the job must be able to:

PC3. Conduct pre-flight inspections using a checklist and identify any potential issues or concerns with drone equipment.

PC4. Apply battery management knowledge to safely handle and maintain drone batteries, demonstrating proper charging, discharging, and storage practices.

System Maintenance and Calibration:

To be competent, the user/individual on the job must be able to:

PC5. Successfully inspect and maintain propulsion systems, including motors, propellers, and ESCs, demonstrating competence in identifying and resolving mechanical issues.

PC6. Perform sensor calibration procedures and troubleshoot flight controller and navigation system issues, ensuring accurate and stable drone operation.

PC7. Demonstrate proficiency in camera and gimbal maintenance by conducting care practices, calibrating equipment, and troubleshooting common issues to ensure optimal imaging quality and stability during flight.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the principles and importance of regular maintenance, safety protocols, and troubleshooting to ensure drone reliability and operational efficiency.

KU2. Gain in-depth knowledge of key drone components, including propulsion systems, flight controllers, navigation systems, batteries, cameras, and gimbals, along with their maintenance and calibration requirements.

KU3. Acquire an understanding of advanced troubleshooting techniques, including flight log analysis, error detection, and addressing emergencies such as power failures, communication loss, and equipment malfunctions.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Develop hands-on proficiency in performing pre-flight inspections, calibrating critical components such as flight controllers, navigation systems, and cameras, and ensuring optimal drone performance.

GS2. Acquire practical skills in diagnosing and resolving issues related to propulsion systems, batteries, and software, as well as handling emergency situations like power loss and communication failures.

GS3. Demonstrate the ability to implement safety guidelines, adhere to regulatory standards, and apply best practices in drone maintenance and operation for reliable and secure performance across various applications.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Drone Maintenance and Troubleshooting:	**11**	**5**	**-**	**-**
PC1. Demonstrate understanding of the importance of maintenance and troubleshooting through written assessments or discussions.	-	-	-	-
PC2. Apply safety considerations and best practices learned to perform maintenance tasks and troubleshoot common drone issues effectively.	-	-	-	-
Equipment Inspection and Maintenance Procedures:	**11**	**5**	**-**	**-**
PC3. Conduct pre-flight inspections using a checklist and identify any potential issues or concerns with drone equipment.	-	-	-	-
PC4. Apply battery management knowledge to safely handle and maintain drone batteries, demonstrating proper charging, discharging, and storage practices.	-	-	-	-
System Maintenance and Calibration:	**11**	**5**	**-**	**-**
PC5. Successfully inspect and maintain propulsion systems, including motors, propellers, and ESCs, demonstrating competence in identifying and resolving mechanical issues.	-	-	-	-
PC6. Perform sensor calibration procedures and troubleshoot flight controller and navigation system issues, ensuring accurate and stable drone operation.	-	-	-	-
PC7. Demonstrate proficiency in camera and gimbal maintenance by conducting care practices, calibrating equipment, and troubleshooting common issues to ensure optimal imaging quality and stability during flight.	-	-	-	-
NOS Total	**33**	**15**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N2506
NOS Name	Drone Equipment Safety and Maintenance
Sector	Electronics
Sub-Sector	
Occupation	Drone Technology
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. identify employability skills required for jobs in various industries

PC2. identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.

PC4. follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC5. recognize the significance of 21st Century Skills for employment

PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC7. use basic English for everyday conversation in different contexts, in person and over the telephone

PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English

PC9. write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

PC10. understand the difference between job and career

PC11. prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings

PC13. work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC14. communicate and behave appropriately with all genders and PwD

PC15. escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC16. select financial institutions, products and services as per requirement

PC17. carry out offline and online financial transactions, safely and securely

PC18. identify common components of salary and compute income, expenses, taxes, investments etc

PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC20. operate digital devices and carry out basic internet operations securely and safely

PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively

PC22. use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research

PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion

PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

PC26. identify different types of customers

PC27. identify and respond to customer requests and needs in a professional manner.

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

GS3. behave politely and appropriately with all
GS4. how to work in a virtual mode
GS5. perform calculations efficiently
GS6. solve problems effectively
GS7. pay attention to details
GS8. manage time efficiently
GS9. maintain hygiene and sanitization to avoid infection

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Employability Skills	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
Constitutional values – Citizenship	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
Basic English Skills	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
Career Development & Goal Setting	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
Communication Skills	**2**	**2**	**-**	**-**
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
Diversity & Inclusion	**1**	**2**	**-**	**-**
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
Financial and Legal Literacy	**2**	**3**	**-**	**-**
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
Essential Digital Skills	**3**	**4**	**-**	**-**
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Entrepreneurship	**2**	**3**	**-**	**-**
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
Customer Service	**1**	**2**	**-**	**-**
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	**2**	**3**	**-**	**-**
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

NIE/ELE/N2509: Implementation of Multi-Functional Drone Systems Using Pixhawk and Python Programming

Description

This project focuses on designing and deploying multi-functional drones using the Pixhawk flight controller and Python programming. It involves integrating various subsystems, including navigation, communication, and payload systems, to achieve seamless operation. Practical applications include drone-based solutions for agriculture, inspection, and environmental monitoring.

Scope

The scope covers the following :

- The project aims to develop versatile drone systems capable of handling diverse real-world applications such as precision agriculture, infrastructure inspection, and disaster management. It includes hands-on experience in subsystem integration, programming, and testing for optimized drone performance. The scope also extends to exploring advancements in drone technology, safety protocols, and regulatory compliance.

Elements and Performance Criteria

Implementation of Multi-Functional Drone Systems Using Pixhawk and Python Programming

To be competent, the user/individual on the job must be able to:

PC1. Understand and identify the components of a drone subsystem, including power systems, flight controllers, propulsion systems, and communication modules, and explain their functions.

PC2. Successfully assemble and integrate all drone subsystems, ensuring proper connections, calibration, and functionality of hardware components like ESCs, GPS, and sensors.

PC3. Demonstrate the ability to troubleshoot integration issues, such as mismatched connections or calibration errors, ensuring smooth operation of the drone.

PC4. Set up a programming environment for Raspberry Pi and Pixhawk, including configuring dependencies and installing necessary software like Dronekit and MAVProxy.

PC5. Develop Python scripts to program the flight controller for basic flight maneuvers, sensor data collection, and autonomous operations.

PC6. Implement and validate specific drone applications (e.g., precision farming, environmental monitoring) by programming and integrating sensors with real-time data processing.

PC7. Perform comprehensive pre-flight inspections, including battery health checks, propeller balancing, and sensor calibration, following established safety guidelines.

PC8. Conduct routine maintenance of critical components such as motors, ESCs, and gimbals, ensuring optimal drone performance and longevity.

PC9. Analyze flight logs and use advanced troubleshooting techniques to diagnose and resolve issues like C2-link loss, power failure, or sensor inaccuracies during operations.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Grasp the calibration techniques and importance of synchronization between various subsystems to ensure precise operations.

KU2. Gain knowledge of Python programming essentials, including flow control, libraries like Dronekit and MAVProxy, and their application in drone operations.

KU3. Familiarize with advanced calibration techniques, software update procedures, and fail-safe mechanisms for ensuring safe and reliable drone operations.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Apply logical reasoning and technical expertise to troubleshoot hardware and software issues in drone operations.

GS2. Effectively communicate technical concepts, system designs, and troubleshooting steps to team members and stakeholders.

GS3. Demonstrate a willingness to explore new tools, frameworks, and techniques, such as AI integration and advanced payload systems, to enhance drone capabilities.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Implementation of Multi-Functional Drone Systems Using Pixhawk and Python Programming	**-**	**-**	**130**	**30**
PC1. Understand and identify the components of a drone subsystem, including power systems, flight controllers, propulsion systems, and communication modules, and explain their functions.	-	-	-	-
PC2. Successfully assemble and integrate all drone subsystems, ensuring proper connections, calibration, and functionality of hardware components like ESCs, GPS, and sensors.	-	-	-	-
PC3. Demonstrate the ability to troubleshoot integration issues, such as mismatched connections or calibration errors, ensuring smooth operation of the drone.	-	-	-	-
PC4. Set up a programming environment for Raspberry Pi and Pixhawk, including configuring dependencies and installing necessary software like Dronekit and MAVProxy.	-	-	-	-
PC5. Develop Python scripts to program the flight controller for basic flight maneuvers, sensor data collection, and autonomous operations.	-	-	-	-
PC6. Implement and validate specific drone applications (e.g., precision farming, environmental monitoring) by programming and integrating sensors with real-time data processing.	-	-	-	-
PC7. Perform comprehensive pre-flight inspections, including battery health checks, propeller balancing, and sensor calibration, following established safety guidelines.	-	-	-	-
PC8. Conduct routine maintenance of critical components such as motors, ESCs, and gimbals, ensuring optimal drone performance and longevity.	-	-	-	-
PC9. Analyze flight logs and use advanced troubleshooting techniques to diagnose and resolve issues like C2-link loss, power failure, or sensor inaccuracies during operations.	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	-	-	130	30

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N2509
NOS Name	Implementation of Multi-Functional Drone Systems Using Pixhawk and Python Programming
Sector	Electronics
Sub-Sector	
Occupation	Drone Technology
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

This section includes the processes involved in identifying, gathering, and interpreting information to evaluate the Candidate on the required competencies of the program.

Assessment of the qualification evaluates candidates to ascertain that they can integrate knowledge, skills and values for carrying out relevant tasks as per the defined learning outcomes and assessment criteria. The underlying principle of assessment is fairness and transparency. The evidence of the outcomes and assessment criteria. Competence acquired by the candidate can be obtained by conducting Theory (Online), Practical assessment, internal assessment, Project/Presentation/ Assignment, Major Project. The emphasis is on the practical demonstration of skills & knowledge gained by the candidate through the training. Each OUTCOME is assessed & marked separately. A candidate is required to pass all OUTCOMES individually based on the passing criteria.

About Examination Pattern:

1. The question papers for the theory exams are set by the Examination wing (assessor) of NIELIT HQS.

2. The assessor assigns the roll number.

3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.

4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.

5. Pass percentage would be 50% marks in each component.

6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).

7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.

8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.

9. The examination will be conducted in English language only.

Quality assurance activities: A pool of questions is created by a subject matter expert and moderated by other SME. Test rules are set beforehand. Random set of questions which are according to syllabus appears which may differ from candidate to candidate. Confidentiality and impartiality are maintained during all the examination and evaluation processes.

Minimum Aggregate Passing % at QP Level : 50

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ELE/N2507.Drone Technology, Regulations & Ethical Considerations-	34	15	0	0	49	10
NIE/ELE/N2508.Drone Subsystem-	33	15	0	0	48	10

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ELE/N2503.Drone Fight Controller & Peripherals	33	15	0	0	48	10
NIE/ELE/N2504.Drone Payload Installation and Integration Testing	33	15	0	0	48	9
NIE/ELE/N2505.Drone Programming and its Applications	34	15	0	0	49	10
NIE/ELE/N2506.Drone Equipment Safety and Maintenance	33	15	0	0	48	9
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
NIE/ELE/N2509.Implementation of Multi-Functional Drone Systems Using Pixhawk and Python Programming	0	0	130	30	160	32
Total	**220**	**120**	**130**	**30**	**500**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AA	Assessment Agency
AB	Awarding Body
NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
National Occupational Standard	NOS define the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service, or technology.